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The Editors

*Bernoulli*

Bernoulli Society for Mathematical Statistics and Probability

Dear Editors,

I am pleased to submit the enclosed manuscript, “*Martingale Innovations from Contractive Recurrent Networks: A Distribution-Free Foundation for Sequential Monitoring*,” for consideration for publication in *Bernoulli*.

The contribution is theoretical and deliberately two-tier. At the first tier, for operator-norm contractive recurrent networks, the hidden state admits a Doob decomposition into a predictable component and a martingale-difference innovation; the transient innovation approaches its stationary counterpart geometrically, and the accumulated innovations obey a functional central limit theorem. A bridge proposition translates this state-level stabilization into a conditional-drift bound for prediction residuals under a Lipschitz readout. At the second tier, for any fitted predictor whose residuals satisfy explicit conditional-drift and sub-Gaussian tail conditions, the Innovation CUSUM enjoys a *sharp* exponential in-control average-run-length lower bound — at the Lundberg rate that coincides with the classical i.i.d. CUSUM exponent, proved by a renewal argument rather than a union bound — together with a logarithmic detection-delay upper bound. The two combine into the exponential-alarm / logarithmic-delay trade-off, which we show is *minimax optimal* in Lorden’s sense, and exactly so under a Gaussian innovation shift; the guarantee is model-free with respect to the data-generating dynamics though not assumption-free. Two further propositions identify when learning is statistically useful: under a nonlinear conditional mean, fixed linear residualization carries an irreducible innovation defect that a consistent learned predictor removes, at a rate that transfers to the monitoring guarantee.

The empirical studies examine these mechanisms rather than claim a performance victory: learned innovations are closer to martingale differences under misspecification, calibration is more stable when the in-control window is short, and equal-ARL comparisons delimit when the learned detector is competitive with correctly specified parametric procedures. The work sits at the interface of probability and mathematical statistics, between the residual-based monitoring tradition (where the model class is specified) and conformal/e-detector approaches (where exchangeability-type conditions replace it), and is, to our knowledge, the first to combine recurrent-state innovation theory with sequential CUSUM calibration for a learned predictor.

In the interest of full transparency, I note that a companion manuscript, “*Backward Coherence and Hidden-State Stability in Recurrent Neural Networks: A Quasi-Reverse-Martingale Theory*” (arXiv:2606.08934), develops a complementary *reverse*-filtration theory for the convergence and stability of the hidden state itself. The present paper works in the *forward* filtration, with the innovations; the two manuscripts are self-contained and neither relies on the other’s results. I would be glad to provide the companion to the editors or referees on request.

The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All proofs, simulation details, and reproducible code are provided in the accompanying Supplementary Material. I have no conflicts of interest to declare.

Thank you for your consideration. I would be happy to provide any additional information the

editors may require.

Sincerely,

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